



Hyperbolic Functions - Worksheet 1

Core Pure 2 Chapter 6: Hyperbolic Functions. Answer all questions. Show your working clearly. Total: 89 marks. Time: 110 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are four sections, one for each topic in Chapter 6. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Give exact answers, using natural logarithms where appropriate. Inverse hyperbolic functions are written $\operatorname{arsinh} x$, $\operatorname{arcosh} x$ and $\operatorname{artanh} x$.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Definitions, graphs and identities	21	
B Solving hyperbolic equations	22	
C Inverse hyperbolic functions and logarithmic forms	21	
D Calculus of hyperbolic functions	20	
Roar-level challenge	5	
Total	89	

Section A Definitions, graphs and identities

21 marks

1. Find the exact values of $\sinh(\ln 3)$ and $\tanh(\ln 3)$. [2]

.....

.....

2. On the same axes, sketch the graphs of $y = \cosh x$ and $y = \sinh x$. State the range of $\cosh x$. [3]



3. Using the definitions of $\cosh x$ and $\sinh x$ in terms of exponentials, prove that

[3]

$$\cosh^2 x - \sinh^2 x \equiv 1.$$

.....

.....

.....

.....

4. (a) Use Osborn's rule to write down the hyperbolic identity that corresponds to $\cos 2x \equiv 1 - 2\sin^2 x$.

[4]

(b) Prove your identity using the exponential definitions.

.....

.....

.....

.....

.....



5. (a) Using the exponential definitions, prove that

[3]

$$\sinh(A + B) \equiv \sinh A \cosh B + \cosh A \sinh B.$$

.....

.....

.....

.....

(b) Hence, using $\sinh 2x \equiv 2 \sinh x \cosh x$ and $\cosh 2x \equiv 1 + 2 \sinh^2 x$, show that

[3]

$$\sinh 3x \equiv 3 \sinh x + 4 \sinh^3 x.$$

.....

.....

.....

.....

(c) Hence solve the equation $\sinh 3x = 11 \sinh x$, giving your answers as exact logarithms.

[3]

.....

.....

.....

.....

(Total for Question 5 is 9 marks)



Section B Solving hyperbolic equations

22 marks

6. Solve the equation $\sinh x = \frac{3}{4}$.

[2]

.....

.....

7. Solve the equation $\cosh x = \frac{5}{4}$.

[3]

.....

.....

.....

.....

8. Solve the equation $5 \cosh x - 3 \sinh x = 5$, giving your answers in exact form.

[4]

.....

.....

.....

.....

.....

9. Using a suitable identity, solve the equation

[4]

$$\cosh 2x - 5 \cosh x + 4 = 0,$$

giving your answers as exact logarithms.

.....

.....

.....

.....

.....



10. (a) Show that the equation $4 \cosh x + \sinh x = k$, where k is a constant, can be written as [3]

$$5e^{2x} - 2ke^x + 3 = 0.$$

.....

.....

.....

.....

- (b) Find the set of values of k for which the equation $4 \cosh x + \sinh x = k$ has real solutions. [3]

.....

.....

.....

.....

- (c) Solve the equation $4 \cosh x + \sinh x = 4$, giving your answers in exact form. [3]

.....

.....

.....

.....

(Total for Question 10 is 9 marks)



Section C Inverse hyperbolic functions and logarithmic forms

21 marks

11. Write $\operatorname{arsinh} 2$ and $\operatorname{arcosh} 3$ as exact natural logarithms. [2]

.....

.....

12. By writing $x = \sinh y$, prove that $\operatorname{arsinh} x = \ln \left(x + \sqrt{x^2 + 1} \right)$. [3]

.....

.....

.....

.....

13. Prove that $\operatorname{artanh} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$, $|x| < 1$. [3]

.....

.....

.....

.....

14. Find the exact value of x for which [4]

$$\operatorname{arcosh} x = \operatorname{arsinh} \left(\frac{3}{4} \right) + \ln 2.$$

.....

.....

.....

.....

.....



15. (a) By writing $x = \cosh y$, where $y \geq 0$, prove that [4]

$$\operatorname{arcosh} x = \ln \left(x + \sqrt{x^2 - 1} \right), \quad x \geq 1,$$

justifying the choice of sign.

.....

.....

.....

.....

.....

- (b) Using the identity $\cosh 2y \equiv 2 \cosh^2 y - 1$, solve the equation $2 \operatorname{arcosh} x = \operatorname{arcosh} 17$. [3]

.....

.....

.....

.....

- (c) Show that $\operatorname{arcosh} 3 = 2 \operatorname{arsinh} 1$. [2]

.....

.....

(Total for Question 15 is 9 marks)



Section D Calculus of hyperbolic functions

20 marks

16. Differentiate with respect to x

[2]

(a) $\cosh 4x$

(b) $x^2 \sinh x$

.....

.....

17. Prove that $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$.

[3]

.....

.....

.....

.....

18. Find $\int \frac{1}{\sqrt{4x^2 - 9}} dx$, $x > \frac{3}{2}$.

[3]

.....

.....

.....

.....

19. By completing the square, show that

[4]

$$\int_1^3 \frac{1}{\sqrt{x^2 - 2x + 5}} dx = \ln(1 + \sqrt{2}).$$

.....

.....

.....

.....

.....



20. (a) Use the substitution $x = 2 \sinh u$ to show that

[5]

$$\int_0^2 \sqrt{x^2 + 4} \, dx = 2\sqrt{2} + 2 \ln(1 + \sqrt{2}).$$

.....

.....

.....

.....

.....

.....

.....

(b) Given that $\frac{d}{dx}(\operatorname{arsinh} x) = \frac{1}{\sqrt{1+x^2}}$, use integration by parts to find $\int \operatorname{arsinh} x \, dx$.

[3]

.....

.....

.....

.....

(Total for Question 20 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

21. The curve C has equation $y = 5 \cosh x - 3 \sinh x$.

[5]

Find the exact coordinates of the stationary point of C , and determine its nature.

.....

.....

.....

.....

.....

.....

.....

.....

.....

END OF WORKSHEET

Total: 89 marks